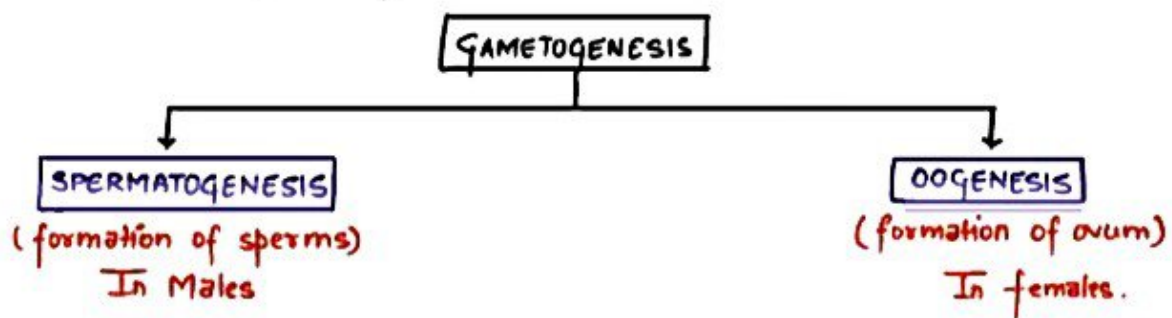


*** REPRODUCTION:-** process of producing offspring, for continuity of generation.

* EVENTS OF REPRODUCTION :-

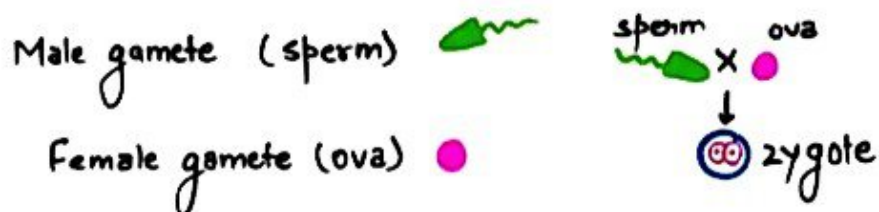
1.) Gametogenesis :- formation of gametes i.e. sperms in → males
ovum in → females.

- It is of two types:-



2) INSEMINATION :- transfer of male gamete (sperm) into the female genital tract.

3) FERTILISATION :- fusion of male and female gametes.



* On the basis of fertilisation animals are of two types :-

1) Oviparous :- Egg-laying animals

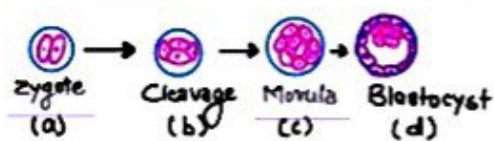
- fertilization is internal but development of embryo is through external means.

2) Viviparous :- Animals giving birth to offspring

- fertilization + development of embryo occur inside the female body.

Humans are sexually reproducing and Viviparous

4) IMPLANTATION :- attachment of blastocyst to the uterine wall of mother.



5) GESTATION :- development of embryo inside uterus (i.e. duration of pregnancy).

6) PARTURITION :- expulsion of fully developed foetus from mother's body.
(i.e. delivery of foetus / child birth).

* Difference in Reproductive Events of Male and Female :-

SPERMATOGENESIS

- Starts at puberty
- Produces motile gametes (sperms)
- Continues in older men

OÖGENESIS

- Starts at embryonic stage
- Produces non-motile gametes.
- formation of ovum ceases in women around the age of 50 years (MENOPAUSE)

Human Reproductive system is made of two types of reproductive or sex organs:-

PRIMARY SEX ORGANS

- Organs responsible for production of gametes and secretion of sex hormones.

Eg.:- Testis → In Male

Ovary - In Female

SECONDARY SEX ORGANS

- Organs do not either produce gamete or secrete sex hormones, but play important role in reproduction, eg:-
transfer, storage, maturation etc.

E.g.:- Epididymis, vas deferens,
seminal vesicle etc. - In Males

Oviduct, uterus, vagina etc. - In female

THE MALE REPRODUCTIVE SYSTEM

* Located in pelvic region

* Male Reproductive System includes -:

→ a pair of testes.

→ Male accessory ducts

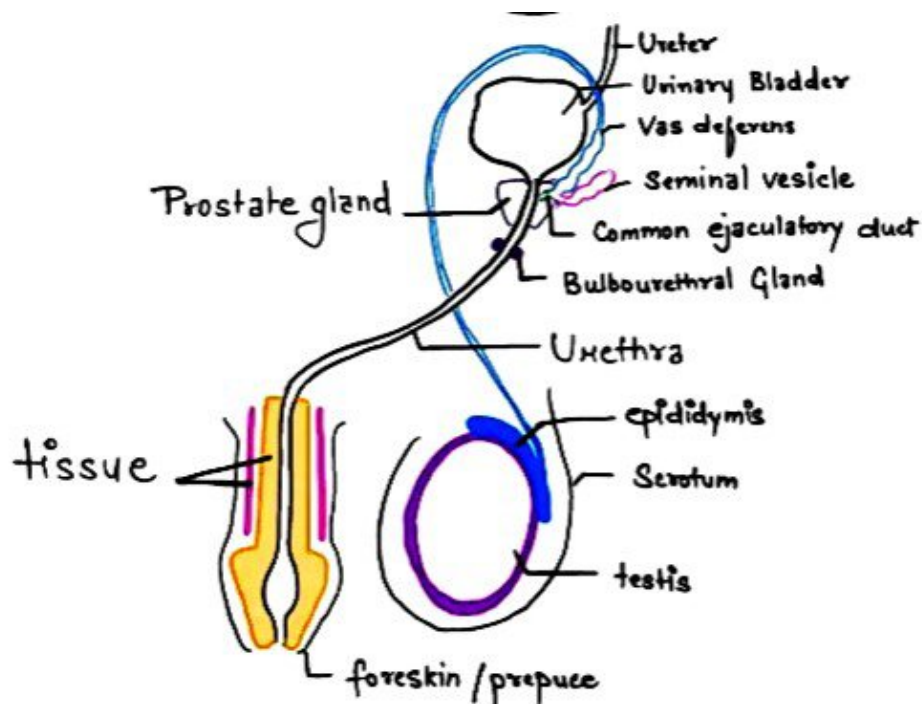
- Rete Testis
- Vasa efferentia
- Epididymis
- Vas deferens.

→ Male accessory glands

- Seminal vesicles
- Prostate gland
- Bulbourethral gland

→ External genitalia

- Penis
- Scrotum

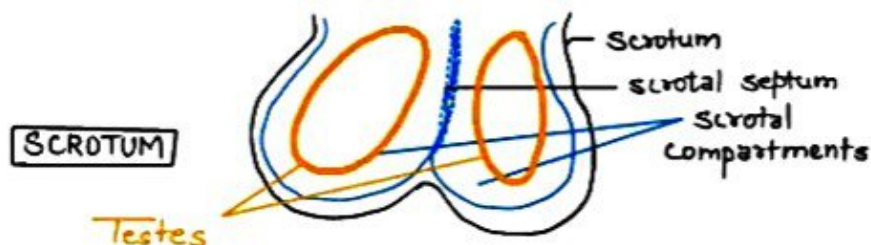


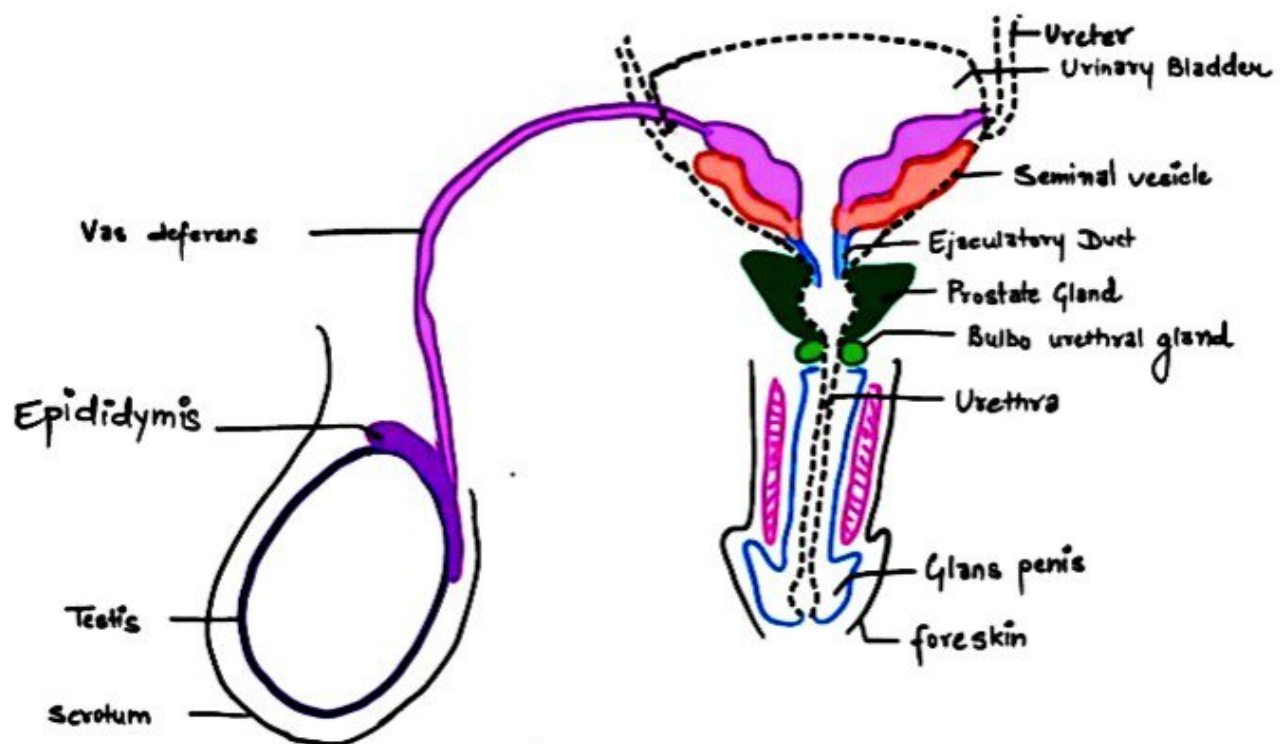
VIEW OF MALE PELVIS

1) SCROTUM:- testes are situated outside abdominal cavity, in a pouch like structure called as scrotum.

•) Helps in maintaining the low temperature of testes ($2-2.5^{\circ}\text{C}$) lower than the body temperature, which is necessary for spermatogenesis.

•) It has two compartments separated by scrotal septum.





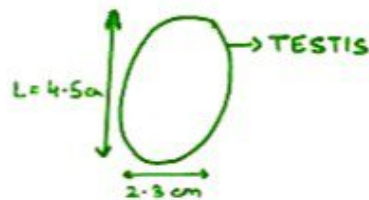
MALE REPRODUCTIVE SYSTEM

2.) TESTES :- paired structure

→ In adults, each testis is oval shaped.

→ Length - 4 to 5 cm

Width - 2 to 3 cm



→ Testis is covered by **three coats** :-

i) Tunica vaginalis :- partially covers testis
has a **parietal** and **visceral** layer.

ii) Tunica albuginea :- it is dense, white fibrous coat.

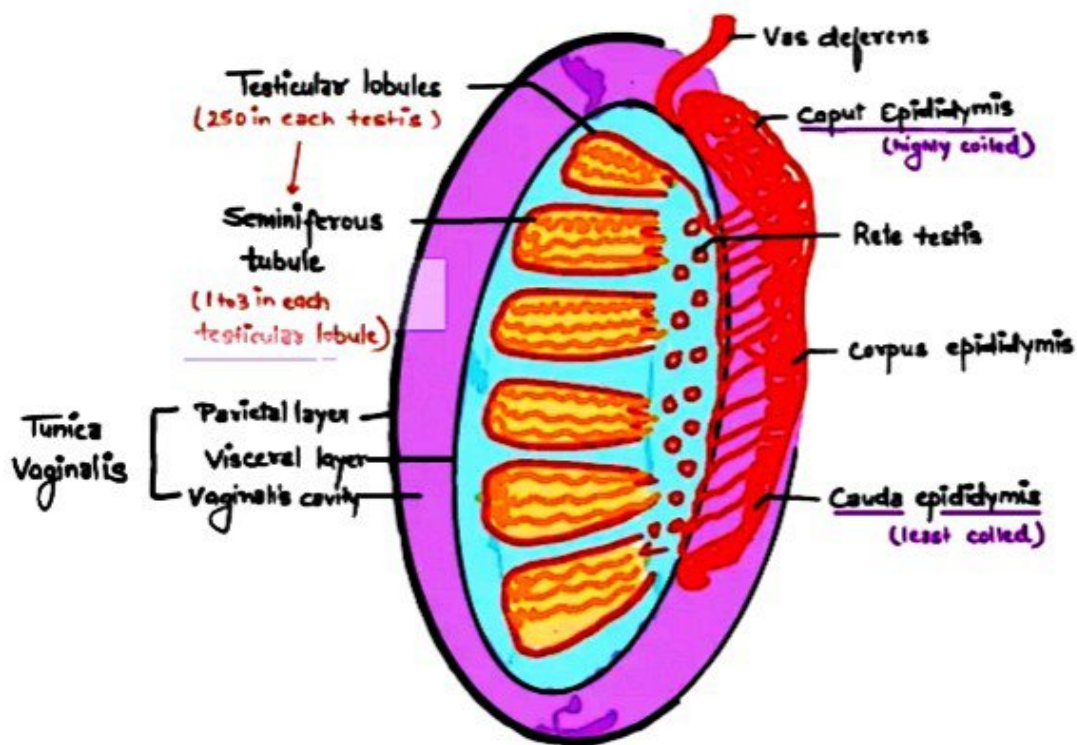
iii) Tunica vasculosa :- vascular coat lining the testicular lobules.

→ Each testis has about **250 compartments** called
testicular lobules.

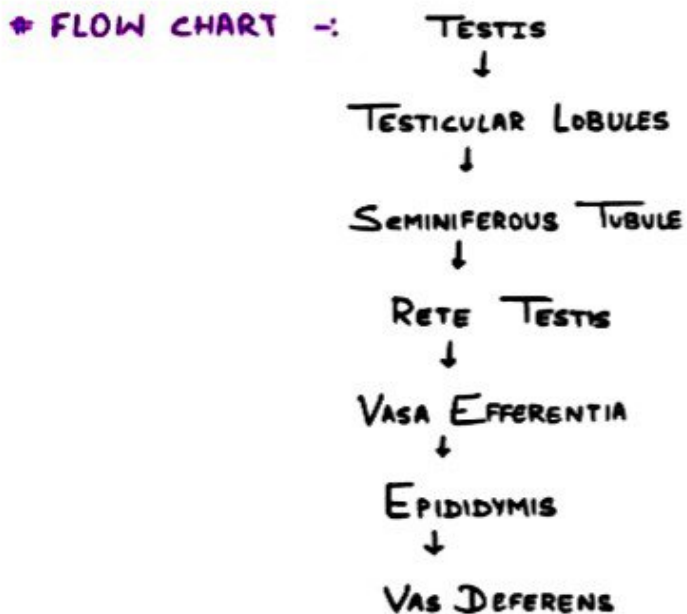
→ Each testicular lobule contains **1-3** highly coiled **seminiferous tubules** in which sperms are produced.

→ Seminiferous tubule opens into vasa efferentia through rete testis.

→ Vasa efferentia leaves the testis, opens into epididymis.



INTERNAL STRUCTURE OF TESTIS



3) EPIDIDYMIS :-

- Located along the posterior surface of testis.
- It is a single convoluted tube, about 6m long.
- Has 3 parts :- Caput epididymis
Corpus epididymis
Cauda epididymis :- opens in vas deferens.

4) VAS DEFERENS :- paired structure

- Epididymis leads to vas deferens.
- It ascends to abdomen and loops over urinary bladder.
- It receives a duct from seminal vesicle and opens into urethra as ejaculatory duct.
- Conveys sperms during sexual arousal.

MALE ACCESSORY GLANDS :-

Includes :- seminal vesicles → paired

3 prostate → unpaired

bulbo urethral glands → paired.

Secretions of these glands constitute the seminal plasma.

(A) SEMINAL VESICLE :- paired

- lie at the dorsolateral sides of base of urinary bladder.
- Each seminal vesicle, joins ampulla of vas deferens to form ejaculatory duct.
- Secretions :- alkaline viscous fluid → rich in fructose (nutrition of sperm) fibrinogen, prostaglandins etc.
- Majority of volume of seminal plasma is secreted by seminal vesicles.

(B) PROSTATE GLAND :- Unpaired

- Lies below urinary bladder
- Prostatic secretions :- alkaline milky contains calcium, clotting enzyme etc.

(C) BULBOURETHRAL GLAND :- also called Cowper's Glands.

- Paired
- Lies at the sides of urethra.

→ Secretions :- alkaline fluid helps in protection of sperms and lubrication of penis.

SEMEN :- Seminal plasma + sperms

→ pH of semen (7.3-7.5) slightly alkaline

→ Human male ejaculates about 200-300 million sperm during a coitus.

→ For normal fertility :- 60% sperms must have normal shape & size.

40% must show vigorous motility.

(5) PENIS :- male external genitalia.

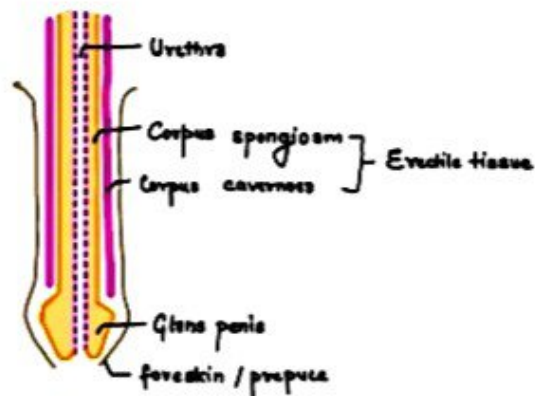
→ Made up of special tissue that helps in erection and facilitate insemination.

→ Enlarged part is called glans penis

Covered by loose fold of skin called as foreskin.

★ Both semen & urine are carried out by penis

EXTERNAL GENITALIA



(6) URETHRA :- originates from urinary bladder.

→ extends through penis, and opens in external opening called urethral meatus.

⇒ PATHWAY OF TRANSPORT OF SPERMS :-

Seminiferous tubule → Rete testis → Vasa efferentia → Epididymis
ejaculatory duct ← Ampulla ← Vas deferens
↳ Urethra → out.

SEMINIFEROUS TUBULE :-

→ Lined on its inside by germinal epithelium.

→ Has two types of cells :-

Male germ cell :- cuboidal shape

→ mature male germ cell is called as spermatogonia.

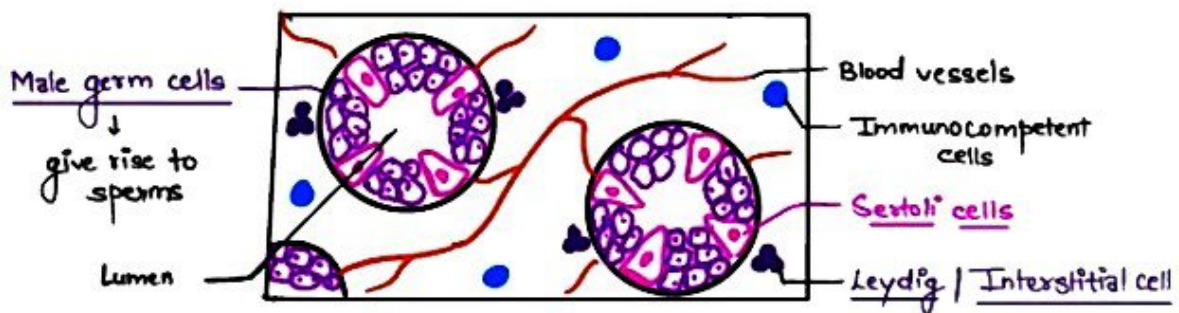
→ Undergoes meiotic divisions, leading to sperm formation.

Sertoli cells :- columnar shaped

- provide nutrition to germ cells.
- phagocytosis of dead, improper sperms.
- Secrete factors called **sperm maturation factor**.
- Also secrete **ABP** - **Androgen Binding Protein** - binds to androgens and helps in its concentration in testis

→ Region outside seminiferous tubule is called interstitial spaces, contains small blood vessels and **leydig cells / Interstitial cells**.

Leydig cells :- synthesise and secrete testicular hormone called androgens (mainly testosterone).



SECTION OF TESTICULAR LOBULE

